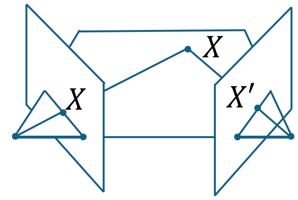
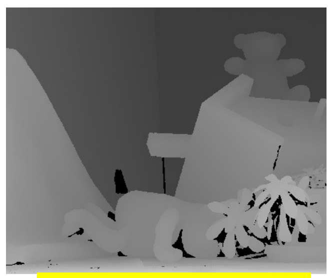
Stereo Vision



- Compute depth from two views

Not Casual: Stereo parameters (R, T) are known with 100% accuracy


Disparity Map

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Stereo Vision

$$Z = \frac{fT}{x_r - x_l} = \frac{fT}{d}$$

- Compute depth from two views
  - ~~When we are given casual pictures, not necessarily from a device such as a Kinect or calibrated cameras~~

In this special case, the pictures are far from casual; very specific cameras




Disparity Map

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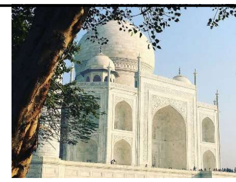
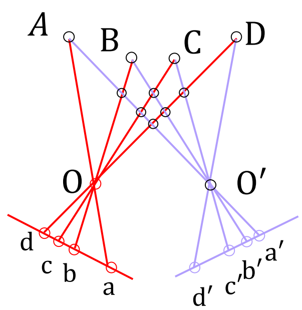
Key Challenge



- Given "casual" left and right images, how can we (efficiently) find the correspondence for random points in the scene?
  - The "fundamental" matrix is an important step
  - (if we are confident on some 7 corresponding points)



Nothing == casual picture

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Overview

- How to do reconstruction if parameters are known
  - Unaligned but known cameras
- Some fun stuff!!
- Basics, buzzwords, and terminology on stereo
- The fundamental matrix **F**
  - Computing 'parameters' from **F**
- The essential matrix **E**
  - Computing 'parameters' from **E**

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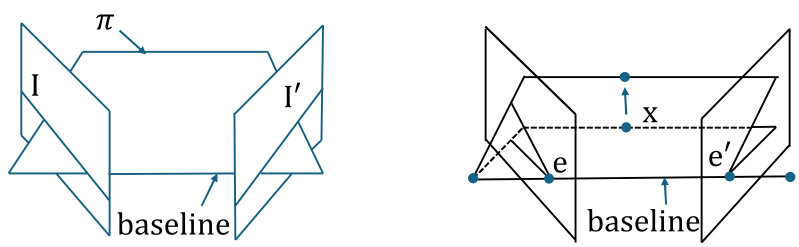
Terminology: Epipolar Geometry

- An **epipolar plane** is a plane containing the baseline
  - There is a one parameter family of epipolar planes
  - The intersection of the epipolar plane with an image plane results in a line

The focus in this slide is on the baseline



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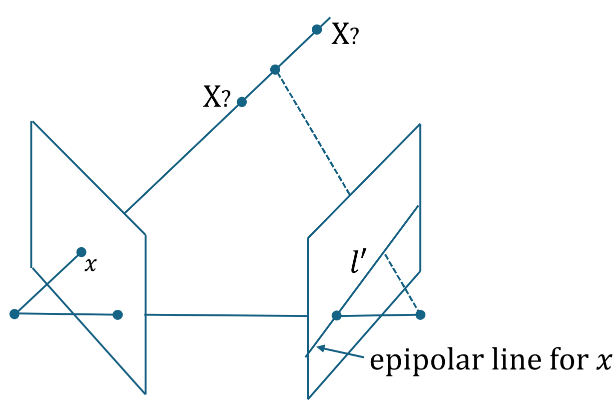
Terminology: Epipolar Geometry

- Point x in the first image results in a mapping to an **epipolar line**

The focus in this slide is on the image point x and the resulting line l'



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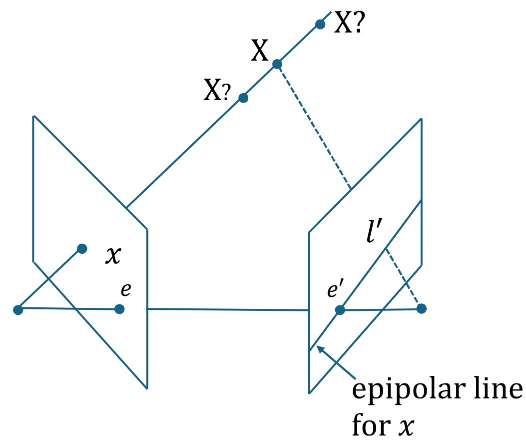
Terminology: Epipolar Geometry

- An epipole is (equivalently)
  - The image of the "other" camera center
  - The vanishing point of the baseline
  - Point of intersection of the baseline with image plane

The focus in this slide is on the two image points ***e*** and ***e'***



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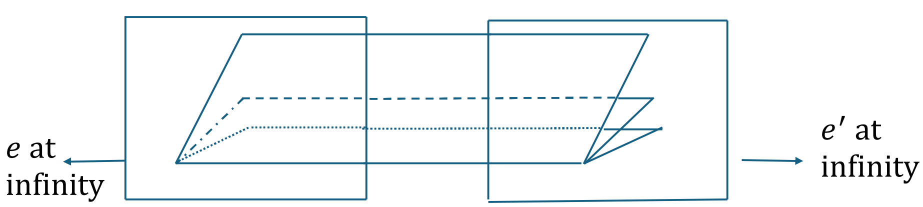
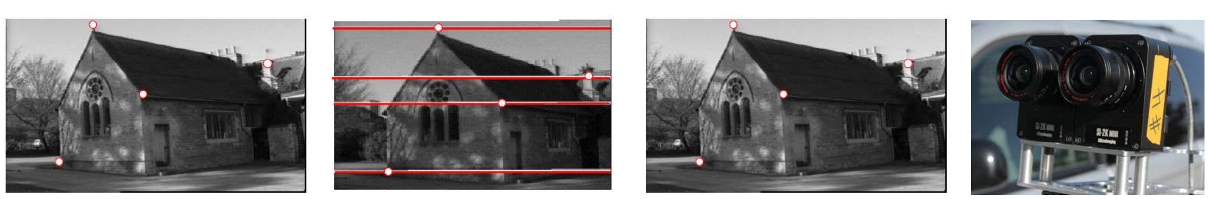
Example: Special case



Where are the epipolar lines and epipoles?



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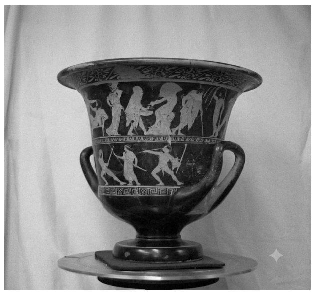
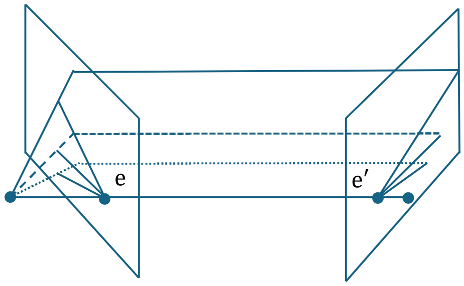
Example: converging cameras





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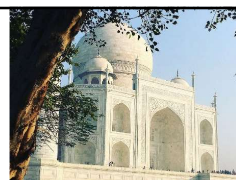
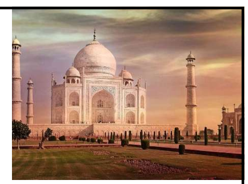
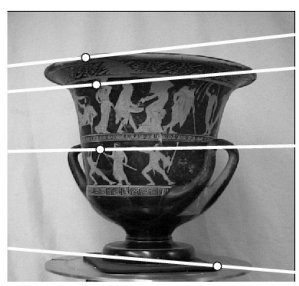
Key Challenge





- Given "casual" left and right images, how can we (efficiently) find the correspondence for random points in the scene?
  - The "fundamental" matrix and the epipolar line is an important step



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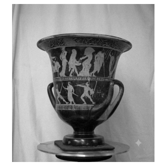
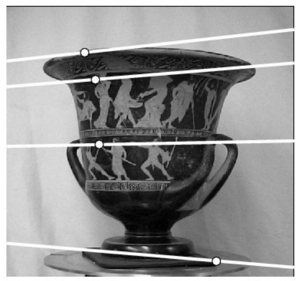
Punch Line: Epipolar lines



- Given two pictures, and a point, one can draw an epipolar line based on an algebraic notion viz. a matrix



The focus in this slide is on the epipolar lines drawn on the image using a matrix eclipsing the 3D geometry

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Punch Line: Epipolar lines

- Given two pictures, and a point, one can draw an epipolar line based on an algebraic notion viz. a matrix

The focus in this slide is on the epipolar lines drawn on the image with a matrix eclipsing the 3D geometry

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Skew-Symmetric Matrix and Cross Product

- If $a = (a_1, a_2, a_3)^T$ define
- Then $a \times b$ is given by the matrix product
- Note the skew-symmetric matrix S
  - Rank is 2
  - $x^T S x = 0$

$$[a]_\times = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} = S$$

$$a \times b = [a]_\times b$$

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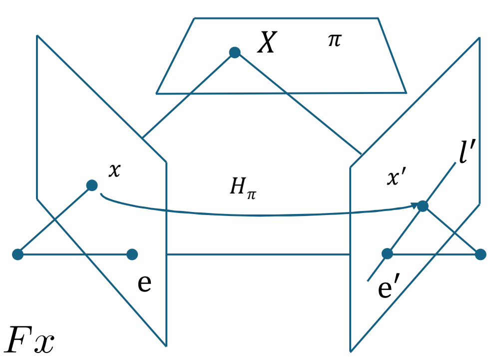
Form of F

We did this before!

- Points x and x' are both images of a point X living on a hypothetical plane π
- The left and right planes are projectively equivalent, thus there is a homography H_π between x and x'
- The line is thus obtained by



$$l' = e' \times x' = e' \times H_\pi x = [e']_\times H_\pi x = Fx$$

If we know F, then we can draw the line using projective geometry

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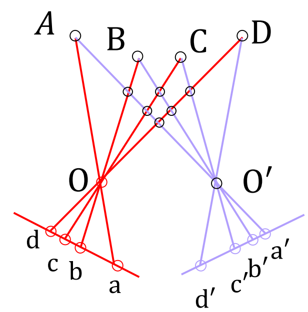
Why F is Fundamental

- Review Key Challenge
- Given “casual” left and right images, how can we (efficiently) find the correspondence for random points in the scene?
  - The “fundamental” matrix is an important step



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Why is F fundamental?

- If x and x’ are corresponding pairs, then x’ lies on the epipolar line
- We can loop over multiple x once we know F to solve correspondence
  - The 2D problem is reduced to a 1D problem

Reminder: The pictures taken from the cameras now are totally casual – we know nothing about them except the image points

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Why is F fundamental?

- If x and x' are corresponding pairs, then x' lies on the epipolar line
  - But the epipolar line is Fx
  - Therefore $x'^T F x = 0$
- Given image points we see how they are fundamentally constrained
- The F matrix is independent of the scene structure, but only dependent on the cameras and the camera arrangement

Reminder: The pictures taken from the cameras now are totally casual – we know nothing about them except the image points

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Overview

- Some fun stuff
- Basics, buzzwords, and terminology on stereo
- The fundamental matrix $\boldsymbol{F}$
  - Computing 'parameters' from $\boldsymbol{F}$
- The essential matrix $\boldsymbol{E}$

The pictures taken from the cameras are totally casual – we know nothing about them except the image points

Despite the casual pictures, we can compute F

Once we get F, it's easier to solve for correspondence

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Determining F

- If x and x' are corresponding pairs, then $x'^T F x = 0$
  - Note the order: If F is the fundamental matrix of cameras (P, P') then F^T is the matrix of the opposite pair
- Flipping the argument,
  - This form characterizes the matrix
  - We can compute this matrix if we know a few corresponding pairs
  - And keep in mind that the rank of the matrix is 2

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Determining F

$(x, y, 1)$ $(x', y', 1)$ $x'^T F x = 0$

$$[x' \; y' \; 1] \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = 0$$

$$[x' \;\; y' \;\; 1] \begin{bmatrix} f_{11}x + f_{12}y + f_{13} \\ f_{21}x + f_{22}y + f_{23} \\ f_{31}x + f_{32}y + f_{33} \end{bmatrix} = 0$$

$$F = \begin{bmatrix} f_{11} \\ f_{12} \\ f_{13} \\ f_{21} \\ f_{22} \\ f_{23} \\ f_{31} \\ f_{32} \\ f_{33} \end{bmatrix}$$

9×1

$$x'xf_{11} + x'yf_{12} + x'f_{13} + y'xf_{21} + y'yf_{22} + y'f_{23} + f_{31}x + f_{32}y + f_{33} = 0$$

$[x'x \;\; x'y \;\; x' \;\; y'x \;\; y'y \;\; y' \;\; x \;\; y \;\; 1]$

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Algorithm for F

$\forall i, 1 \leq i \leq N, \tilde{p}_{r,i}^T [F] \tilde{p}_{l,i} = 0$

Let $\tilde{p}_{r,i} = (x_{r,i}, y_{r,i}, 1), \tilde{p}_{l,i} = (x_{l,i}, y_{l,i}, 1)$

$$\begin{bmatrix} x_{r,1}x_{l,1} & x_{r,1}y_{l,1} & x_{r,1} & y_{r,1}x_{l,1} & y_{r,1}y_{l,1} & y_{r,1} & x_{l,1} & y_{l,1} & 1 \\ x_{r,2}x_{l,2} & x_{r,2}y_{l,2} & x_{r,2} & y_{r,2}x_{l,2} & y_{r,2}y_{l,2} & y_{r,2} & x_{l,2} & y_{l,2} & 1 \\ . & . & . & . & . & . & . & . & . \\ . & . & . & . & . & . & . & . & . \\ x_{r,N}x_{l,N} & x_{r,N}y_{l,N} & x_{r,N} & y_{r,N}x_{l,N} & y_{r,N}y_{l,N} & y_{r,N} & x_{l,N} & y_{l,N} & 1 \end{bmatrix} \begin{bmatrix} F_{11} \\ F_{12} \\ F_{13} \\ F_{21} \\ F_{22} \\ F_{23} \\ F_{31} \\ F_{32} \\ F_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

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Determining F

- The fundamental matrix $\boldsymbol{F}$ has 7 DOF
  - The first two rows $=\ 6\ DOF$
  - Plus the third row = linear combination of first two rows, giving 8 DOF
  - Minus 1 since F is homogeneous
- We can compute F from 7 paired points
- There exist algorithms that need only 7 points, but they are not as simple as the 8-point algorithm using DLT
  - Note: these 8 pairs can be obtained from manual input or using a feature mapping algorithm

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Eight point algorithm

- We solve for $\boldsymbol{f}$ (which contains the 9 entries of $\boldsymbol{F}$) by computing the SVD of $\boldsymbol{A}$ (size N by 9, N ≥ 8) and taking the column vector from $\boldsymbol{V}$ corresponding to the least singular value
  - Ideally $\boldsymbol{A}$ has rank 8 (given $\boldsymbol{f}$ is homogeneous) but in practice $\boldsymbol{A}$ has rank 9 (due to errors in measurement of point coordinates)
  - The solution is obtained up to an arbitrary sign and scaling constant
- $\boldsymbol{F}$ has size 3 by 3, but it should have rank 2, i.e., it should be rank-deficient. The previous step does not guarantee rank-deficiency.
  - So we need another step

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Eight point algorithm

- Rearrange elements of $\boldsymbol{f}$ to give $\boldsymbol{F}$ (up to a scaling constant)
- Compute SVD of $\boldsymbol{F}$ and nullify its smallest singular value. This gives us the final $\boldsymbol{F}$.

$\boldsymbol{U_F S_F V_F^T} = \boldsymbol{F}$

Let $\mathrm{S_F} = \mathrm{diag}(a, b, c), a \geq b \geq c$

$\boldsymbol{F}_{\text{final}} = \boldsymbol{U}_{\mathrm{F}}\mathrm{diag}(a, b, 0)\boldsymbol{V_F^T}$

Find the nearest rank-2 matrix!

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Eight point algorithm

- Rearrange elements of $\boldsymbol{f}$ to give $\boldsymbol{F}$ (up to a scaling constant)
- Compute SVD of $\boldsymbol{F}$ and nullify its smallest singular value. This gives us the final $\boldsymbol{F}$
- Wait, we are not done yet!

$\boldsymbol{U_F S_F V_F^T} = \boldsymbol{F}$
Let $\mathrm{S_F} = \mathrm{diag}(a, b, c), a \geq b \geq c$
$\boldsymbol{F}_{\mathrm{final}} = \boldsymbol{U}_{\mathrm{F}}\mathrm{diag}(a, b, 0)\boldsymbol{V_F^T}$

Find the nearest rank-2 matrix!

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Normalization

- Without normalization image points such as (1000,1000,1) can cause havoc in the numerical computation
- Very important for all DLT like methods such as Homography, Camera, and Fundamental Matrix computation

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Modified Eight point algorithm

$* \ \bar{x}_r = \frac{\sum_{i=1}^{N} x_{r,i}}{N}, \bar{y}_r = \frac{\sum_{i=1}^{N} y_{r,i}}{N}, \bar{x}_l = \frac{\sum_{i=1}^{N} x_{l,i}}{N}, \bar{y}_l = \frac{\sum_{i=1}^{N} y_{l,i}}{N}$

$* \ \sigma_r = \frac{\sum_{i=1}^{N} \sqrt{(x_{r,i} - \bar{x}_r)^2 + (y_{r,i} - \bar{y}_r)^2}}{N}, \sigma_l$
$= \frac{\sum_{i=1}^{N} \sqrt{(x_{l,i} - \bar{x}_l)^2 + (y_{l,i} - \bar{y}_l)^2}}{N}$

$* \ x'_{r,i} \leftarrow \frac{x_{r,i} - \bar{x}_r}{\sigma_r}, y'_{r,i} \leftarrow \frac{y_{r,i} - \bar{y}_r}{\sigma_r}, x'_{l,i} \leftarrow \frac{x_{l,i} - \bar{x}_l}{\sigma_l}, y'_{l,i} \leftarrow \frac{y_{l,i} - \bar{y}_l}{\sigma_l}$

$*$ Now estimate the fundamental matrix F_1 from $\{(x'_{r,i}, y'_{r,i}), (x'_{l,i}, y'_{l,i})\}_{i=1}^{N}$.

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Eight point algorithm

- Normalize x_l and x_r by T and T'
- Compute F_1
- Denormalize to get $F = T'^T F_1 T$

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